

# Computational Requirements

## 1. Purpose

This document lists the computational requirements to reproduce the data processing, model training, and mapping pipeline used in this project (rainfall raster processing, class-imbalance handling, machine learning model training/tuning, and hazard/vulnerability/risk mapping).

## 2. Hardware Requirements

### 2.1 Reference configuration (used to generate published results)

| Component        | Specification                                 |
|------------------|-----------------------------------------------|
| Machine          | HP Pavilion 15 laptop                         |
| Processor        | Intel Core i5                                 |
| RAM              | 16 GB DDR4, 3200 MHz                          |
| Storage          | NVMe SSD                                      |
| Operating System | Windows 11                                    |
| GPU              | Not required, all experiments run on CPU only |

### 2.2 Minimum recommended specification

| Component | Minimum                                      | Recommended                                                                                                              |
|-----------|----------------------------------------------|--------------------------------------------------------------------------------------------------------------------------|
| Processor | Quad-core (Intel i5 / Ryzen 5 or equivalent) | 6+ core (Intel i7/i9, Ryzen 7/9)                                                                                         |
| RAM       | 8 GB                                         | 16 GB or more (GridSearchCV and raster stacking are memory-intensive)                                                    |
| Storage   | 20 GB free (SSD preferred)                   | 50+ GB free SSD — raw NetCDF, daily GeoTIFF stacks, and per-year rasters accumulate quickly                              |
| GPU       | Not required                                 | Not required (XGBoost/CatBoost/LightGBM run CPU-only in this project; a CUDA-capable GPU can optionally accelerate them) |
| Display   | 1366×768                                     | 1920×1080 or higher (for QGIS raster/map inspection)                                                                     |

#### Notes:

- GridSearchCV hyperparameter tuning (Random Forest, Extra Trees, AdaBoost) is CPU- and RAM-intensive and can take from seconds to several minutes per model depending on core count.
- Processing 30 years of daily NetCDF rainfall data into per-day GeoTIFFs, yearly rasters, and a 30-year composite is a storage intensive task. Therefore, an SSD is strongly recommended over an HDD.

### 3. Software Requirements

#### 3.1 Operating System

Windows 10/11, Linux (Ubuntu 20.04+), or macOS.

Scripts were developed and tested on Windows 11. All libraries used are cross-platform.

#### 3.2 Core Runtime

| Software    | Version used / recommended | Purpose                                                   |
|-------------|----------------------------|-----------------------------------------------------------|
| Python      | 3.10                       | Core scripting language for the entire pipeline           |
| pip / conda | Latest                     | Package management (conda recommended for GDAL/geopandas) |

#### 3.3 GIS Software

| Software                            | Purpose                                                                                                                                                          |
|-------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| QGIS (v3.40 or later recommended)   | Raster layer visualization, symbology, and layout for susceptibility, hazard, and risk maps; used for map display and manual raster inspection during processing |
| GDAL (command-line, incl. gdalwarp) | Reprojection, resampling, and clipping of rainfall and other rasters; called directly via subprocess from the Python scripts                                     |

GDAL is a system-level library/binary — install via conda-forge or an OS package manager (not solely via pip) before installing the Python packages below.

#### 3.4 Development Environment

- Visual Studio Code (used for original development) or any Python IDE / editor (Spyder, PyCharm, Jupyter)
- Scripts use "# %%" cell markers compatible with VS Code and Spyder.

#### 3.5 Python Libraries

| Library            | Purpose                                                      |
|--------------------|--------------------------------------------------------------|
| numpy, pandas      | Numerical computation and tabular data handling              |
| geopandas, shapely | Vector geospatial data handling                              |
| rasterio           | Raster I/O, sampling, and reprojection                       |
| xarray             | Reading and processing NetCDF rainfall data                  |
| jenks              | Jenks natural breaks classification                          |
| scikit-learn       | Train/test split, GridSearchCV, evaluation metrics           |
| imbalanced-learn   | SMOTE oversampling, Tomek Links undersampling                |
| matplotlib         | Plotting (ROC curves, feature importance, comparison charts) |
| shap               | SHAP-based model explainability / feature importance         |

A consolidated requirements.txt is provided in the project repository.